ENGINEERING PORTFOLIO — 2026

Irsyad Ghibran, S.T.

Process Engineer | Extractive Metallurgist | Laboratory Analyst

Surabaya, Indonesia

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01 About me

Metallurgical Engineering graduate of UPN "Veteran" Yogyakarta with research and internship experience in extractive metallurgy.

I bring strong analytical skills, precision, and a willingness to keep learning and adapting, so I can contribute to the metallurgical industry.

FOCUS AREAS

Process Engineering | Extractive Metallurgy | Process Analysis | Laboratory Analysis

EDUCATION

UPN "Veteran" Yogyakarta

Bachelor of Metallurgical Engineering | GPA 3.xx / 4.00

Faculty	Faculty of Mineral Technology
Focus	Extractive Metallurgy
Courses	Extractive Metallurgy, Hydrometallurgy, Pyrometallurgy, Mineral Processing, Physical Metallurgy, Thermodynamics of Materials, Corrosion Engineering
Thesis	Effect of Sulfuric Acid Concentration and Temperature on Nickel Recovery from Laterite Ore by Atmospheric Leaching

02 Technical skills

Extractive Metallurgy

- Hydrometallurgy
- Pyrometallurgy
- Leaching
- Mineral Processing

Laboratory Analysis

- XRD
- XRF
- SEM-EDS
- AAS

Software

- Python
- Spreadsheet Modelling

Process Engineering

- Mass & Energy Balance
- Process Analysis
- Quality Control
- Technical Reporting

03 Work experience

2025

Internship — Extractive Metallurgy

Company name (example)

Example entry. Describe the process unit you worked on, the analyses you ran, and the result of your work.

03 Organization & volunteer

2024

Laboratory Assistant

Metallurgy Laboratory, UPN "Veteran" Yogyakarta (example)

Example entry. Describe the practicums you assisted and the instruments you handled.

04 Projects & research

RESEARCH 2025

Atmospheric Leaching of Nickel Laterite

Hydrometallurgy · Final research

Studied the effect of acid concentration, temperature, and time on nickel recovery from laterite ore, with metal content measured by AAS.

Leaching · Nickel · AAS

01

RESEARCH 2024

Ore Characterization with XRD & XRF

Mineralogy · Laboratory analysis

Identified the mineral phases and elemental composition of ore samples to recommend a suitable extraction route.

XRD · XRF · Mineralogy

02

INTERNSHIP 2025

Smelter Process Efficiency Review

Internship project

Reviewed furnace operating data to find losses in energy use and metal recovery, and proposed practical improvements.

Pyrometallurgy · Process Data

03

CASE STUDY 2024

Mass Balance of a Copper Concentrate Circuit

Process analysis

Built a mass balance model of a flotation circuit to trace copper losses between stages and test what-if scenarios.

Mass Balance · Copper · Spreadsheet

04

05 Case studies

FINAL RESEARCH · HYDROMETALLURGY LAB · 2025

Raising Nickel Recovery in Atmospheric Leaching

Leaching · Design of Experiments · AAS

Challenge

Nickel recovery from low-grade laterite stayed below 60% at standard leaching conditions, making the route uneconomical.

Approach

Ran a factorial experiment on acid concentration, temperature and time, measured dissolved Ni with AAS, and checked residues with XRD.

Result

Identified the condition window that lifted recovery to about 85% (example figure) while limiting iron co-dissolution.

CASE STUDY · PROCESS ANALYSIS · 2024

Tracing Copper Losses in a Flotation Circuit

Mass Balance · Flotation

Challenge

Plant data showed copper recovery dropping without a clear source between rougher, cleaner and scavenger stages.

Approach

Reconciled assay and flow data into a stage-by-stage mass balance and compared scenarios in a spreadsheet model.

Result

Located the largest loss at the scavenger tailings and proposed a regrind step (example outcome).

06 Publications

2025 — **Effect of Sulfuric Acid Concentration on Nickel Laterite Leaching (example)** — JOURNAL

I. Ghibran, Co-author · Journal name (example)

2024 — **Ore Characterization Using XRD and XRF for Process Selection (example)** — CONFERENCE

I. Ghibran · Seminar / conference name (example)

07 Certificates

Certificate image

Certificate name 1
Issuer

Certificate image

Certificate name 2
Issuer

Certificate image

Certificate name 3
Issuer

08 — CONTACT

Let's get in touch.

Open to opportunities in Process Engineering, Extractive Metallurgy, Process Analysis.